

Interplay of Gyroid Structures for Butterflies in Flight

1105513

April 2026 - Total Words 3877

1 Introduction

The flight mechanisms of a butterfly wing have been long under studies [Buchanan, 2023], the theory of aerodynamics for the flapping of the wing via a reduction energy yields a optimal mannerism to partake in flight[Srygley, 2007], such models have been in study in order to be replicated into to the constructions of aircraft's, we also have a body of researchers in china, who are replicating such mechanisms in robotics [Huang et al., 2023]. Gyroid structures found within many intracellular membrane structures have been a point of discussion, namely their formation, and why these partiucular structures appear. There formation has been hypothesised to be through the polymerization of chitin, yet how the particular minimal surface evolves, have been studied through the use of x-ray microscopy and imaging of chitin nano structures [Wilts et al., 2017]. We interplay with the optimisation features of the Gyroid structures during butterfly flight. We shall first purpose by formulating the problem, and thereby answering the questions formulated.

2 Motivation and Background details

We begin by proposing the problem, and motivating factors. The existence of the Gyroid structures at a wing scale (micro-meters) have been proven via microscopy techniques [Wilts et al., 2017]. Given the sensitivity of the preservation of a minimal surface lies on its local curvature at each neighboring point, we shall explore how the minimal surfaces evolves in flight. Namely, the Gyroid structure studied are in the reference state, however it is well known that muscle activation induces a change in the intrinsic curvature of an elephant trunk [Kaczmariski et al., 2024], it has been shown using experimental biology techniques [Kammer, 1967] that the butterfly also contracts its wings via fibre activation at a microscale, therefore a natural question be, given the intrinsic curvature of a butterfly wing changes in flight, and in fact at every point in flight for all times t , how does the minimal surface evolve, is it still a Gyroid structure, or is it no longer a minimal surface, or is it a a transition phase between the two,

i.e. above a critical yield stress generated through Kinetic and Potential energy in the butterfly it transitions between the two states. Given the minimal surface at hand be a triply periodic structure, with gaps in its structure, how may that assist in the flight mechanisms of a butterfly. All these are questions which we delve on answering. We shall first propose the generic definitions to navigate through the project. After which, we shall present a model to geometrically describe the wing, thereafter we present a local argument to prove what occur at the interphase region of a the wing indeed be a minimal surface. Thereafter the plan is to derive a PDE for the Gyroid structure via free energy principles, and lastly to couple the PDE's for the Geometry and the Gyroid, to see how the two interplay, and what what we be interested in, namely its coupling in flight.

3 Mathematical Definitions/key tools

3.1 Geometry of surfaces

Let Σ be an orient-able surface with parametrization $\mathbf{x}(\xi^1, \xi^2) \in \mathbb{R}^3$ where $(\xi^1, \xi^2) \in M \subset \mathbb{R}^2$ - Assume $\mathbf{x}$ at least C^2 such that $\mathbf{r}_i = \frac{\partial \mathbf{x}}{\partial \xi^i}$ are linearly independent $\forall (\xi^1, \xi^2) \in M$. We can then define the unit normal $\mathbf{n} = \frac{\mathbf{r}_1 \wedge \mathbf{r}_2}{|\mathbf{r}_1 \wedge \mathbf{r}_2|}$, noting that $(\mathbf{r}_1, \mathbf{r}_2, \mathbf{n})$ forms a basis.

Surface area: $A = \int_{\Sigma} dS$, $dS = |\mathbf{r}_1 \wedge \mathbf{r}_2| d\xi^1 d\xi^2$
Which after employing the identity: $|\mathbf{r}_1 \wedge \mathbf{r}_2|^2 = r_1^2 r_2^2 - (\mathbf{r}_1 \cdot \mathbf{r}_2)^2$,
, and further using the definition of Metric tensor:

$g_{ij} = \mathbf{r}_i \cdot \mathbf{r}_j$ we may write the the above as

$$dS = \sqrt{g_{11}g_{22} - g_{12}^2} d\xi^1 d\xi^2 \text{ which is the determinant in index notation}$$

$$dS = \sqrt{\det(g)} d\xi^1 d\xi^2$$

Hence, our surface area over the the 2d space M in which my parameters (ξ^1, ξ^2) lie, may be written as: $A = \int_M \sqrt{\det(g)} d\xi^1 d\xi^2$

, and the arc length for any arbitrary path in my surface Σ , between two arbitrary end points also closed with in Σ , can be calculated to be: $L = \int_a^b \sqrt{g_{ij} \frac{d\xi^i}{dt} \frac{d\xi^j}{dt}} dt$, where $ds^2 = g_{ij} d\xi^i d\xi^j$, which is most famously known as the **first fundamental form**.

Next we derive an expression for the extrinsic curvature, by way of first defining the normal curvature. Consider a curve C on Σ passing through a point P and parametrized by arc length s . Take $\mathbf{t}(s) = \frac{d\mathbf{x}}{ds}$ to be the tangent vector of C at P and $k = \left| \frac{d\mathbf{t}}{ds} \right|$ to be the curvature of C . We may **decompose** the curvature $k = k_n + k_g$, where k_n is the normals curvature of C , given by $k_n = \mathbf{k} \cdot \mathbf{n}$. Define $K_{ij} = -\mathbf{n} \cdot \frac{\partial \mathbf{r}_i}{\partial \xi^j}$ which be the extrinsic curvature tensor, note it is symmetric and left as a minor exercise to confirm. Then we can say $k_n = K_{ij} \frac{d\xi^i}{ds} \frac{d\xi^j}{ds}$ where $(K_{ij} d\xi^i d\xi^j)$ is the **second fundamental form**.

We calculate extremal values of k_n , which results in the equation: $(K_{ij} - k_n g_{ij}) \frac{d\xi^i}{ds} \frac{d\xi^j}{ds} = 0$ which is derived by equating the second fundamental form to k_n multiplied by a portion of arc length (ds^2) . Note K_{ij} and g_{ij} will depend on the surface and not the path C. This yields the eigen value problem: $G^{-1}K \mathbf{a} = k_n \mathbf{a}$ where $\mathbf{a}_i := \frac{d\xi^i}{ds}$ then note $k_n = k_n(a_1, a_2)$. Our principle curvatures k_i are eigen values of the principle curvature matrix.

We can therefore define the Mean curvature and gaussian curvature of a surface, for the interest of our 2 dimensional case, we use the following definitions:

$$H = \frac{k_1 + k_2}{2}$$

$K_G = k_1 k_2$ note the gaussian curvature is intrinsic to the surface, it only depends on the metric and not on the normal vector. As a summary we say, we define a minimal surface to be a surface with mean curvature being zero, that is the sum of the principle curvatures, which namely be the eigen values of the product of the the extrinsic curvature matrix K_{ij} and the inverse of the metric tensor matrix G_{ij} . The formulation used to derive the principle curvatures relies on two main definitions:

Our metric tensor $g_{ij} = \mathbf{r}_i \cdot \mathbf{r}_j = \frac{\partial \mathbf{x}}{\partial \xi^i} \cdot \frac{\partial \mathbf{x}}{\partial \xi^j}$ where for an orientable surface Σ the parametrization is given by $\mathbf{x}(\xi^1, \xi^2) \in \mathbb{R}^3$, whence $G = (g_{ij})$ be the matrix of metric tensor.

The extrinsic curvature tensor as $K_{ij} = -\mathbf{n} \cdot \frac{\partial \mathbf{r}_i}{\partial \xi^j} = K_{ji}$ (It be symmetric), this allows us to define $K = (K_{ij})$ to be our matrix of extrinsic curvature tensor.

We will now derive a co-ordinate free form for the mean curvature. Consider a surface as a graph $z = h(x, y) \in C^2$. Firstly given the De-Monge representation of a surface $\mathbf{r} = (h(x), h(y), h(x, y))^T$

Then we may write: $\mathbf{r}(x, y) = (x, y, h(x, y))$ taking deirvatives with respect to the independant variables, yield $\mathbf{r}_x = (1, 0, h_x)$ and $\mathbf{r}_y = (0, 1, h_y)$ and the $\det(G) = g = 1 + h_x^2 + h_y^2$ Our normal vector: $\mathbf{n} = \frac{(-h_x, -h_y, 1)}{\sqrt{1+h_x^2+h_y^2}}$.

The extrinsic curvature tensor matrix and the Mean curvature:

$$K = \frac{1}{\sqrt{g}} \begin{pmatrix} h_{xx} & h_{xy} \\ h_{xy} & h_{yy} \end{pmatrix}$$

$$H = \frac{1}{2} \text{Tr}(G^{-1}K) \text{ which is } 2H = \nabla \cdot \left(\frac{\nabla h}{\sqrt{g}} \right)$$

Note $2H = \nabla \cdot \left(\frac{\nabla h}{\sqrt{g}} \right)$ which is the same as $\nabla(\nabla \cdot \mathbf{n})$ hence we yield the the condition for a minimal surface in co-ordinate free form, namely

$$H = 0 \implies \nabla \cdot \left(\frac{\nabla h}{\sqrt{g}} \right) = 0 \implies \nabla \cdot \mathbf{n} = 0 \quad (1)$$

4 Geometry

In this section we will model the geometrical construction of the butterfly wing and state appropriate assumptions made.

Geometrical non linearity (Large or small deformations) Assumptions:

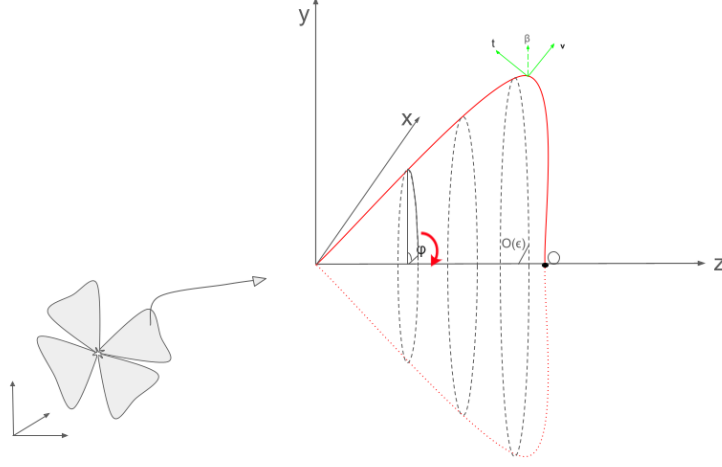


Figure 1: A part of a wing geometrically modelled as an axi-symmetric membrane with a $\mathcal{O}(\epsilon)$ thickness in the azimuthal direction. With a material frame along the arc length of the wing to be the basis: (τ, v, β)

1. Assume small deformations of the wing from the reference state to the current state (i.e. flying) though this may be an assumption we drop later if at the wing scale the deformations be large.
2. We employ the model for axisymmetry to model the butterfly wing, though we start with the model of the membrane (only resisting bending but not stretching) and pursuit to expand on the shells model where we consider both bending and stretching

Consider the wing to be an extensible axisymmetric membrane filled with an incompressible fluid [Dallmann and Herman, 1978], we will consider the fluid to be distributed homogeneously in the wing, and we further assume that the concentration remains constant in flight and rest. We take one of the wing, from either the forewings or the Hind wings, at this moment we assume their geometry to be the same. An objective later be to will dwell on this to give a rigorous treatment of the forces acting on the forewings as compared to the Hind wings, further how the energy profiles differ in either case.

For the geometry we will be mainly using the membrane model [Goriely, 2017]. To form the wing we rotate a planar curve C about the z -axis as shown in figure 1

The initial configuration i.e. when the butterfly be at rest, be characterised by: $\underline{r}(S, t) = r(S, t)\underline{e}_r + z(S, t)\underline{e}_z, S \in [0, L]$ where L be the total length around the wing before any deformation. We further will parametrise in terms of $r(S, t)$ and $\theta(S, t)$.

After using the symmetrical properties of the (τ, v, β) frame, we arrive at a

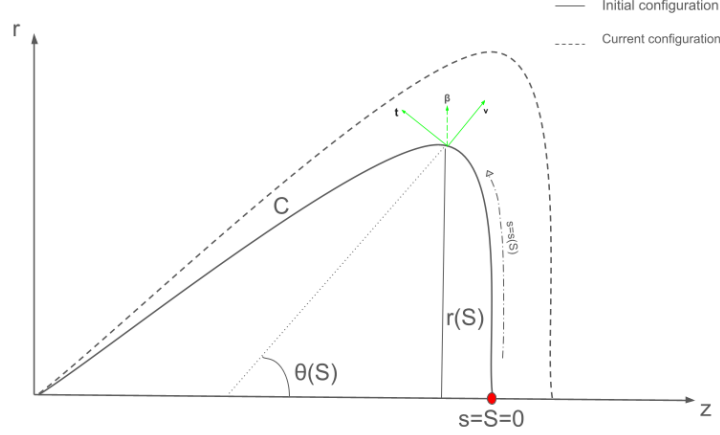


Figure 2: A material point $s(S)$ is measured from the Apex of the wing, and its position be $r(S)$ on the curve C , the angle θ be the angle between the normal (τ) and the z -axis

relationship for the tangent (τ) and normal (v), namely,

$$\underline{\tau} = \cos(\theta)\underline{e}_r + \sin(\theta)\underline{e}_z \text{ and } \underline{v} = \sin(\theta)\underline{e}_r + \cos(\theta)\underline{e}_z$$

from which using the relation $\underline{\beta} = \underline{\tau} \times \underline{v}$ one may derive the binormal vector note here and in the diagram the binormal vector is pointing into the board.

4.1 principle curvatures

To make use of the definitions given earlier, we first parametrize our surface:

$$\underline{X} = \begin{pmatrix} r(s)\cos(\phi) \\ \epsilon r(s)\sin(\phi) \\ z(s) \end{pmatrix}$$

We next calculate the extremals of $L = G^{-1}K$ as per our definitions and diagonalise, yielding. Note that this deals with the singularity in our boundary conditions namely at $\theta = n\pi, n \in \mathbb{Z}$

$$k_s = \frac{\epsilon}{\sqrt{\epsilon^2 + (1 - \epsilon^2)\sin^2\phi\sin^2\theta}}$$

$$k_\phi = -\frac{\epsilon}{(\epsilon^2 + (1 - \epsilon^2)\sin^2\phi\sin^2\theta)^{\frac{3}{2}}}$$

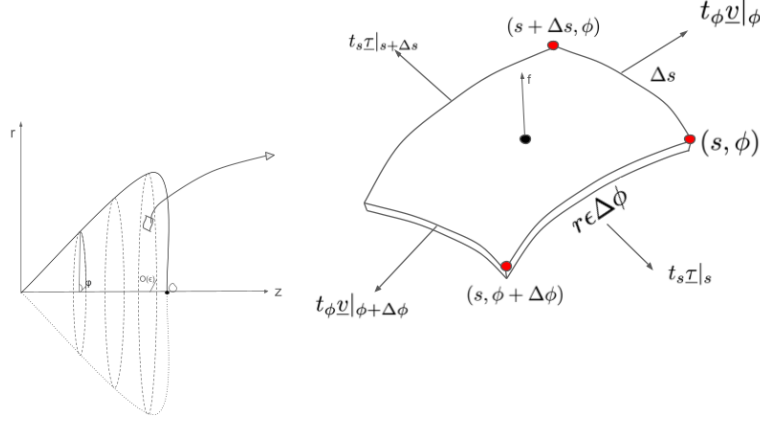


Figure 3: The respective line tensions corresponding to the portion of the wing on a microscale

5 Mechanics

In this section we will consider the forces acting on the wing, and will derive the dynamical equations through a force balance procedure.

We consider a portion of the wing as illustrated in figure 3, and consider the forces acting upon the piece.

We define:

t_s to be the line tension across tangent with units $\frac{Force}{Length}$

t_ϕ - tension along the binormal (β)

$\underline{f}$ - we consider this to be the external body force, this be a vector and we later write explicitly. One choice may be $\underline{f} = \tau_c \underline{\tau} + \underline{P} \cdot \underline{v}$

$\underline{\tau}$ be the shear stress exerted on the wing, note as the butterfly uses its potential energy to flap the wing, it also thrusts the forces exerted on it, i.e. gravity and air pressure.

P be the pressure difference along the wing

5.1 Force Balance

We next consider the force balance, note need be made that the forces are acting on the section per unit area, therefore we will commute the forces with the area to yield the complete force balance equation. The full force balance based in the schematic diagram fig3:

$$[(t_s \underline{\tau} \epsilon r)|_{s+\Delta s} - [(t_s \underline{\tau} \epsilon r)|_s] d\phi + [(t_\phi \beta)|_{\phi+\Delta\phi} - (t_\phi \beta)|_\phi] ds + \epsilon \underline{f} r ds d\phi = 0 \quad (2)$$

taking $ds \rightarrow 0$ and $d\phi \rightarrow 0$, we equate the respective tangential, normal and binormal components.

β :

$$\frac{\partial t_\phi}{\partial \phi} = 0 \quad (3)$$

no dependence on ϕ required for axisymmetry, therefore the non trivial components are the normal and tangential components, note here we consider the formulation of $\underline{f} = \tau_c \underline{\tau} + \underline{P} \cdot \underline{v}$, though we may drop this to be identically zero for our resting state:

Force in the Normal and tangential direction be:

$$0 = \frac{\epsilon^2}{\sqrt{\epsilon^2 + (1 - \epsilon^2)\sin^2\phi\sin^2\theta}} rt_s + -\frac{\epsilon}{(\epsilon^2 + (1 - \epsilon^2)\sin^2\phi\sin^2\theta)^{\frac{3}{2}}} t_\phi - \epsilon r P \quad (4)$$

$$\epsilon \frac{\partial}{\partial s}(rt_s) - t_\phi \cos\theta - r\epsilon\tau_c = 0 \quad (5)$$

Deformations:

Figure 2 shows the later deformation of the wing, though it need not be precise, as the factors one need consider are the deformation in each respective direction, it is completely possible and valid that the deformation only occurs near a local region at the apex, or it may totally be valid for the deformation to contract rather than expand, i.e. the butterfly has larger wings in resting state than in flying state. Each of these cases and complications can further be considered and exploited, and we shall try doing so given the scope of the project at hand. Furthermore, solutions to such equations generally referred to as the membrane equations in the literature, have had many approaches to finding solutions, one may see [Pipkin, 1968] where the absence of shear stress and constant pressure be considered.

we define the stretches of the wing shape which allow us to categorise the deformation:

$$\lambda_s = \frac{\partial s(S,t)}{\partial S}$$

$$\lambda_\phi = \frac{r(S,t)}{R(S)}$$

5.2 Constitutive relations

Here we discuss Material non linearity (stress and strain relation) namely how do the stress and strain relation ship relate, we may assume in the beginning that the applied stress becomes easier and later harder i.e. hookean by considering the wing to be an elastic like material. In relating the tensions to the stretches, we first purpose the most simplest model, that be the neo-hookean relation, for our strain energy density function $W = \frac{A}{2}(I_1 - 3)$ where I_1 is the first scalar principle invariant of a tensor say $\mathbf{A}$, which means it be the solution to the characteristic equation $\det(\mathbf{A} - \lambda_i \mathbf{I}) = 0$, $I_1 = \text{tr} \mathbf{A} = \sum_{i=1}^3 \lambda_i$, where λ_i be the principle stretches of the wing under deformation. Note there's a third stretch, namely acting in the normal direction of the wing λ_n , but we

will assume material incompressibility (or the isochoric condition), therefore $\lambda_n = 1/\lambda_s\lambda_\phi$. Our tensions be in terms of the principle stretches therefore, it be most appropriate to use the following relation $t_i = J^{-1}\lambda_i \frac{\partial W}{\partial \lambda_i} - p\mathbf{1}$. Note we set the constraint (acting as a Lagrange multiplier) of the pressure, namely the $-p\mathbf{1}$ term to 0, to only consider the effect of gravity, to explore further, we may employ an isotropic constitutive law of the form :

$$t_s = t_\phi = A(\lambda_s^2 + \lambda_\phi^2 - 2) \quad (6)$$

Such that $t_s = t_\phi = F(\lambda_s, \lambda_\phi) = F(1, 1) = 0$, but that mainly be used for spherical geometry, and we have differing tensions at every point locally on the surface of the wing. We will shortly see that, further creativity be required to define the tensions at the apex of the wing, to yield the desired shape.

6 Closing the System and Simple Solutions

For the case of our external forces being in terms of gravity only, we write: $f = -\rho g \underline{v}$

$$\frac{\partial t_s}{\partial S} = \frac{\lambda_s}{r} \cos(\theta) \left[\frac{t_\phi}{\epsilon} - t_s \right] \quad (7)$$

$$\theta' r t_s - \frac{\epsilon}{(\epsilon^2 + (1 - \epsilon^2) \sin^2 \phi \sin^2 \theta)^{\frac{3}{2}}} t_\phi + \epsilon r \rho g = 0 \quad (8)$$

$$\frac{\partial r}{\partial S} = \lambda_s \cos(\theta) \quad (9)$$

$$\frac{\partial z}{\partial S} = -\lambda_s \sin(\theta) \quad (10)$$

$$t_s = J^{-1} \lambda_s \frac{\partial W}{\partial \lambda_s} - p\mathbf{1} = A\lambda_s^2 \quad (11)$$

$$t_\phi = J^{-1} \lambda_\phi \frac{\partial W}{\partial \lambda_\phi} - p\mathbf{1} = A\lambda_\phi^2 \quad (12)$$

Where we have assumed $k_s \approx \frac{\partial \theta}{\partial S}$, note this be true if we had an isotropically distributed spherical geometry, though this we do but locally, also because we have only taken k_s to be at leading order of ϵ we must ensure the $\approx$ sign. note A be the material parameter for the wing, namely the Youngs Modulus. We now have 4 Differential equations, and 2 algebraic equations relating the tension to the stretches therefore our system be closed. We now define boundary conditions and plot the wing for the most simplest case namely only gravity effect. We have also converted our force balance equations from before in terms of the later state s to now in terms of S our current state.

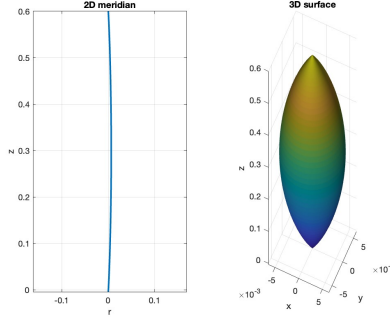


Figure 4: $A = 0.01 \text{ GPa}$, $\rho = 1$, $g \approx 9.8 \text{ m/s}^2$, $L = 2(\text{cm})$, $a = 0.6(\text{cm})$, $\phi = \pi/2$, $\delta = 0.05$, $\epsilon = 0.65$

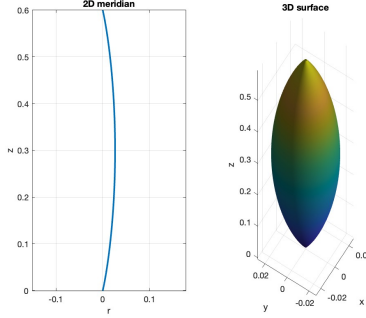
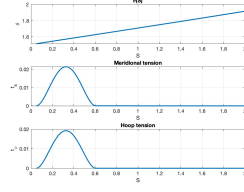


Figure 5: $A = 1.1 \text{ GPa}$, $\rho = 1$, $g \approx 9.8 \text{ m/s}^2$, $L = 2(\text{cm})$, $a = 0.6(\text{cm})$, $\phi = \pi/2$, $\delta = 0.05$, $\epsilon = 0.5$

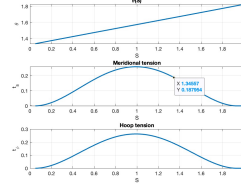
6.1 Boundary Conditions

Based on fig2 we take the boundary condition $\theta(0) = \frac{\pi}{2}$, $\theta(L/2) = \frac{\pi}{2}$, $z(S=0) = a$, $r(S=0) = 0$. We must note that $r(0) = 0$ yields a singularity in our equation (6) therefore we use a backward Taylor expansion to perturb the solution near a parameter $S = \delta$, hence yielding; $r(0) = r(\delta) - r'(\delta)\delta + O(\delta^2)$ and similarly we do this for z and θ , therefore our boundary conditions will now take the form: $\theta(\delta) = \frac{\pi}{2}$, $\theta(L/2) = \frac{\pi}{2}$, $z(\delta) = a$, $r(\delta) = 0$ where $\delta \ll 1$. We plot the results below.

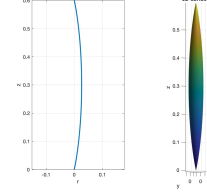
Figure 4 has been plotted using the Youngs modulus close to that of rubber, not very stiff except above a particular threshold, whereas figure 5 has been plotted using approximately the Youngs modulus of Polytetrafluoroethylene (PTFE), molded, mainly used in making pans, to prevent non stickiness to the surface of the surface, this Youngs modulus has produced a better approximation of the 2D meridian curve, therefore we speculate some properties



(a) tension plot corresponding to figure 4



(b) tension plot corresponding to figure 4



(c) Side elevation for figure 5

of Polytetrafluoroethylene (PTFE) may be found at the wing surface, and helps in some functionality as expected. We also note that there be a deficiency of an extra condition that allows the modelling of differing tensions across the apex and across the centrlne of the wing. We require some creativity to cause the apex region to be more turgent, similar to figure 1. However, this suffices for a local analysis. Though, we will propose the model for bio-plates be used instead [Goriely, 2017], or some other boundary condition be imposed around the apex.

7 Construction of hypersurfaces via limits of the Allen-Cahn solutions

The main result we'll be exploiting in this section, will be quoted much later, namely that due to [Pacard and Ritoré, 2003]. Concisely, If we have a minimal surface in a manifold, we can obtain it as a limit of the Allen Cahn energy extremal. We first derive the Allen-Cahn Equation by first principles, to understand the system, and the assumptions embedded therein. We then link the appropriate assumptions to the surface of the butterfly wing, under study.

The Allen-Cahn equation is a member of a larger group of equations, namely the reaction diffusion equations, which be used to describe the phase compositions between mixtures with in a system. To motivate the equation we first consider the composition of two mixtures, namely our alpha and beta compositions x_B^α , x_B^β , if we are to look at the schematic representation in 7. We look at going from $\alpha \rightarrow \beta$, we observe that there's a "sudden change" between the two mixtures, we call that "sudden change" the plane of discontinuity or the interface region. If one is to assume that α **and** β are uniform throughout the phase, then at the interface there's a free energy (this is the distinguishing factor in this phase field equation, namely how it differs from the general diffusion equation). If we are to have a continuous curve instead (for the interfacial region), the gradient of that curve will contribute to the free energy and need be accounted for. For the schematic plot 7 we may see that only the compositions contribute to the free energy of our system, however we highlight two problems with proceeding with such a schematic representation, in deriving our phase field equation.

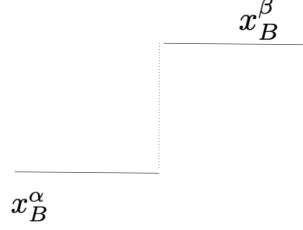


Figure 7: A schematic plot of the alpha and beta composition, with the dashed line representing our "sudden change" or the interphase between the two compositions.

1. We assume a constant α and β composition, and the interface has a discontinuity.
2. System not thermodynamically acceptable, as once the thermodynamic parameters are set, no other information needed, the system will decide which interface to be taken

Hence we model our free energy for the system, considering the two problems above. Let F be a function of the independent variable $\in \mathfrak{R}^3$, and all its order derivatives, therefore curvatures. We may first write the following.

$$F(x, y, z, c, \frac{\partial c}{\partial x}, \frac{\partial c}{\partial y}, \frac{\partial c}{\partial z}, \frac{\partial^2 c}{\partial x^2}, \frac{\partial^2 c}{\partial x \partial y}, \frac{\partial^2 c}{\partial x \partial z} \dots) = \int f dV \quad (13)$$

Where f is our free energy, we look at a modification of our free energy by means of a Taylor expansion:

$$f = f(c) + \frac{\partial f}{\partial \nabla c} (\nabla c)_i + \frac{\partial f}{\partial (\nabla^2 c)_{ij}} : (\nabla^2 c)_{ij} + \frac{1}{2!} \frac{\partial^2 f}{\partial (\nabla c)_i \partial (\nabla c)_j} : (\nabla c)_i (\nabla c)_j + \dots$$

Which we may write as:

$$f = f(c) + \alpha_i (\nabla c)_i + \beta_{ij} (\nabla^2 c)_{ij} + \gamma_{ij} (\nabla c)_i (\nabla c)_j + \dots \quad (14)$$

Noting that α_i is a first rank tensor and β_{ij} and γ_{ij} be 2nd rank tensors. We have also assumed Einstein's summation convention. In our expansion we have considered the nominal composition of c , where c has small changes, note that if there are no changes, then the free energy is the first term in the expansion.

Assume that we are in a system, where there's centre or in motion symmetry (inversion symmetry), this be at the scale of the wing. Therefore, all odd rank tensors be zero, namely $\alpha \equiv 0$ and we expand and put the remaining terms in the same form:

$$\int \beta_{ij} (\nabla^2 c)_{ij} dV = \int \beta_{ij} \frac{\partial^2 c}{\partial x_i \partial x_j} dV = \int \beta_{ij} \frac{\partial}{\partial x_i} \left(\frac{\partial c}{\partial x_j} \right) dV$$

$$= \int \beta_{ij} \frac{\partial}{\partial x_i} \left(\frac{\partial c}{\partial x_j} \right) dV = \int \beta_{ij} \frac{\partial}{\partial x_j} (\nabla c)_j = \beta_j (\nabla c)_j |_S - \int \frac{\partial \beta_{ij}}{\partial x_i} (\nabla c)_j dV$$

Where in the last step we carried out, integration by parts. We will choose our surface S for the volume V in such a way that the opposite surfaces on the wing will give equal contributions, hence it will be zero. We plug therefore this result as part of the Taylor expansion into our free energy integral 13, yielding:

$$F = \int [f(c) + \left(\frac{\gamma_{ij}}{2} - \frac{\partial \beta_{ij}}{\partial c} \right) (\nabla c)_i (\nabla c)_j] dV = \int [f(c) + K_{ij} (\nabla c)_i (\nabla c)_j] dV \quad (15)$$

Where we let $\frac{\gamma_{ij}}{2} - \frac{\partial \beta_{ij}}{\partial c} = K_{ij}$. We further assume cubic systems for our wing, hence $K_{ij} = K_{ji}$, and in such systems the order of $(\nabla c)_i (\nabla c)_j$ wouldn't matter, such tensors will be isotropic, hence:

$$F = \int f(c) + K \delta_{ij} (\nabla c)_i (\nabla c)_j dV = \int [f(c) + K (\nabla^2 c)] dV \quad (16)$$

Here recall $c = c(x, y, z, t)$ and ∇c is the gradient of c therefore we can use the Euler Lagrange equation.

$$\frac{\delta F}{\delta c} = \frac{\partial f}{\partial c} - \nabla \cdot 2K (\nabla c) = \frac{\partial f}{\partial c} - 2K \nabla^2 c = N\mu \quad (17)$$

Where we have assumed K to be constant, we have also made use of the well known result in the last line, namely that $\frac{1}{N} \frac{\delta F}{\delta c} = \mu$ where N is the number of atoms in the system, and μ is the chemical potential. We in the last line, express the chemical potential in the case of a functional.

Before the free energy in the wing (system) was only a function of the composition c , now also a function of the composition gradient, that is because we want to incorporate changes from the interphase, which is done by incorporating the gradient. We remind our selves of the relation between the Diffusive flux and the chemical potential:

$$J = -M \nabla \mu \implies \frac{\partial c}{\partial t} = -\nabla J = M \nabla^2 \mu \quad (18)$$

Given we take M to be constant. Pluggig in into our μ -fuctional representation 17:

$$\frac{1}{N} \frac{\partial c}{\partial t} = M \nabla^2 \left(\frac{\partial f}{\partial c} - 2K \nabla^2 c \right) = M \nabla^2 \left(\frac{\partial f}{\partial c} \right) - 2KM \nabla^4 c \quad (19)$$

Writing in terms of per atom, we get rid of the factor $1/N$, therefore yield

$$\boxed{\frac{\partial c}{\partial t} = M \nabla^2 \left(\frac{\partial f}{\partial c} \right) - 2KM \nabla^4 c} \quad (20)$$

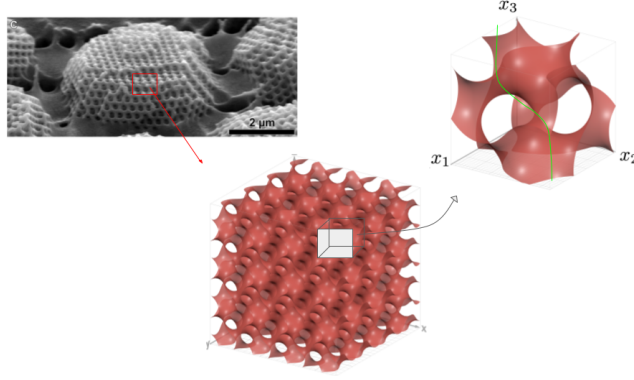


Figure 8: A schematic diagram, of us locally parameterising the Gyroid structure. Note the microscopic picture labelled c has been taken from [Wilts et al., 2017]

This is the Cahn-Hilliard equation, a modified diffusion equation. The modified term in K came from the concentration gradient and the concentration gradient represents difference, hence this term is related to interface, which will then relate to the energy effects. Interface will now choose the width for the competition between two compositions[[Shames, 2017]] We now re-label our variables as we shift from the literature of spinodal decompositions and phase field equations, to the literature surrounding minimal surfaces coupled with the Allen-cahn equation. Note here, f is our double well potential.

The above equation may be written as

$$\frac{\partial u}{\partial t} = M \nabla^2 \left(\frac{\partial f}{\partial u} \right) - 2KM \nabla^4 u = \nabla^2 (u^3 - u) - \epsilon \nabla^4 u \quad (21)$$

Now spot that that factorising the negative Laplacian of the right hand side of our equation ??, yields the Allen-cahn equation:

$$\frac{\partial u}{\partial t} = \nabla^2 (u^3 - u) - \epsilon \nabla^4 u = -\nabla^2 (u - u^3 + \nabla^2 u) \quad (22)$$

Now assuming there to be no mass conservation in our system we may get rid of the Laplacian [Cahn and Hilliard, 1959], or simply scale it to be 1 hence yielding the desired result. In one way the negative Laplacian operator induces a mass conservation effect into our Cahn Hilliard system. Now our equation becomes.

$$\boxed{\frac{\partial u}{\partial t} = u - u^3 + \nabla^2 u} \quad (23)$$

Which is the Allen-Cahn Equation.

Theorem - Pacard-Ritoré Given $\Sigma \subset M$ a smooth, separating, non-degenerate minimal hypersurface, $\exists \epsilon_0 > 0$, such that $\forall \epsilon \in (0, \epsilon_0) \exists u$ a solution to (25) such that $\{u = 0\} \rightarrow \Sigma$ as a smooth graph [Pacard and Ritoré, 2003].

We consider a Sub Manifold (M, g) , where the free energy $E_\epsilon : C^\infty(M) \rightarrow \mathbb{R}$, for the Allen-Cahn Equation, the free energy functional is of the form:

$$E_\epsilon[u] = \int_M \frac{\epsilon}{2} |\nabla u|^2 + \frac{W(u)}{\epsilon} d\mu_g \quad (24)$$

Where $W : \mathbb{R} \rightarrow \mathbb{R}$, $W(t) = \frac{1}{4}(1 - t^2)^2$ which be our standard double potential (note this be a choice and other potentials may be chosen depending on the problem, but we here exploit the symmetry of this such case about the point which be degenerate.).

We next deform our energy E_ϵ about some free parameter lets say η , whence yielding our Allen-Cahn energy extremal:

$$\frac{W'(u)}{\epsilon} - \epsilon \nabla^2 u = 0 \quad (25)$$

The solutions (u_ϵ) of which in the limit of $\epsilon \rightarrow 0$ converges in a Hausdorff manner to a minimal surface as per [Pacard and Ritoré, 2003]. For the double - Potential $W(u) = \frac{1}{4}(1 - u^2)^2$, in the 1D case, we yield the solution for the interface region.

$$u_0(t) = \tanh\left(\frac{t}{\epsilon\sqrt{2}}\right) \quad (26)$$

Where in the limit for $\epsilon \rightarrow 0$ will approach to the most simplest minimal surface, namely a point. Note this will be the shape our transition layer takes for the compositions α and β as discussed earlier, and the degenerate region (dotted line) in fig7

How this links to our case is the following, we know that u_0 be a solution to the energy function, therefore in the $\lim \epsilon \rightarrow 0, u_0$ must converge in a Hausdorff manner to a minimal surface, note that our solutions of weight $u_\epsilon = \pm 1$ "concentrates" towards a minimal surface also. We want to show that on the wing, the Gyroid also be a solution to the Allen-Cahn extremum equation. We do this (locally) by focusing on a patch say P in our wings surface, where $P \in \mathbb{R}^3$. We define local co-ordinates within that patch (x_1, x_2) and the signed distance x_3 acting in the normal direction $x_3 \bar{n}$, this forms our parameterization of the patch within the confined region P by a way of two tangential co-ordinates (x_1, x_2) and a normal direction $x_3 \bar{n}$, we define $f : (x_1, x_2) \rightarrow P$, namely the function $f(x_1, x_2)$ which takes us to the Gyroid. A schematic representation of this be in 8

Next let

$$u(f(x_1, x_2) + x_3 \bar{n}) = u_0(x_3) \quad (27)$$

Taking the Laplacian of (27) and plugging into (25):

$$\nabla u(f(x_1, x_2) + x_3 \bar{n}) = \bar{n} \partial u_{x_3} + \partial_{\text{tangential terms}} = \bar{n} \partial u_{x_3} \quad (28)$$

$$\nabla(\bar{n} \partial u_{x_3}) = \nabla \cdot \bar{n} (\partial u_{x_3}) + \partial u_{x_3 x_3} = \nabla^2 u \quad (29)$$

using the result (1) to write our H mean curvature in co-ordinate free form, and plugging our result into (??)

$$\frac{u^3 - u}{\epsilon} - \epsilon(\nabla \cdot \bar{n} (\partial u_{x_3}) + \partial u_{x_3 x_3}) = 0 \implies \frac{u^3 - u}{\epsilon} - \epsilon \partial u_{x_3 x_3} = \epsilon(H(\partial u_{x_3})) \quad (30)$$

Which we see is approximately a solution of the Allen-Cahn extremal by $O(\epsilon)$.

Allen–Cahn approximation near a minimal surface: the Gyroid

The Gyroid structures can be trigonometrically approximated as:

$$F(x, y, z) = \sin x \cos y + \sin y \cos z + \sin z \cos x \quad (31)$$

and define the Gyroid surface as the level set $\Sigma = \{x \in \mathbb{R}^3 : F(x) = 0\}$ To connect this with the Gyroid, we instead define an ansatz using the level set function:

$$u_\epsilon(x_1, x_2, x_3) = u_0 \left(\frac{F(x_1, x_2, x_3)}{\epsilon} \right). \quad (32)$$

so this construction is consistent with the local formulation. We compute:

$$\nabla u_\epsilon = \frac{u'_0}{\epsilon} \nabla F,$$

$$\nabla^2 u_\epsilon = \frac{u''_0}{\epsilon^2} |\nabla F|^2 + \frac{u'_0}{\epsilon} \Delta F.$$

Substituting into the Allen–Cahn equation gives

$$\frac{u_0^3 - u_0}{\epsilon} - \frac{u''_0}{\epsilon} |\nabla F|^2 - u'_0 \Delta F = 0.$$

Using $u''_0 = u_0^3 - u_0$, we obtain

$$\frac{u_0^3 - u_0}{\epsilon} (1 - |\nabla F|^2) - u'_0 \Delta F = 0.$$

For the Gyroid, one computes

$$\Delta F = -2F.$$

Thus, on the surface $F = 0$, we have $\Delta F = 0$, and near the interface $F = O(\epsilon)$ implies $\Delta F = O(\epsilon)$. Therefore the residual of the Allen–Cahn equation is of order $O(\epsilon)$.

Hence, the Gyroid defines an interface along which the Allen–Cahn equation is satisfied up to $O(\epsilon)$, and in the sharp interface limit $\epsilon \rightarrow 0$.

8 Mechanics of the Gyroid Structure

The following work [Jones et al., 2023] allows us to convince ourselves, why this particular structure is found at the wings, and how it potentially be advantageous for the flight. Butterflies need minimal mass and enough stiffness to resist air-loads, Gyroid structures provide high stiffness to weight ratio, this also supports our geometrical model, where we yielded better results for a higher Youngs Modulus. Surface Gyroids are shown to also be stretch dominated, which implies that stretching membrane tension is more efficient than bending, which means the butterfly needs less structural deformation during flapping. It also has fluid and structural advantages, as Gyroids are co-continuous structures, which is shown to be useful for fluid transport in the wings (though we assumed a homogeneous distribution, but this complication may be considered for a more sophisticated model.). the property of being co-continuous also allows for a more efficient stress distribution across the wing. All of which be associated to our main factors of consideration in the modelling aspect of the project.

9 Acknowledgments and final remarks

This work requires a lot of areas of continuing work, namely to drop major assumptions in the model, to derive a PDE for the Gyroid structure via some free energy principles, a plate model consideration rather than the membrane model from [Goriely, 2017], to couple the mechanics, geometry and Gyroid PDE to explore the wing in flight, under more than the forces considered for the most simplest model above. Further points of improvement have been mentioned alongside the assumptions in the paper. We shall end by the quote from Sir Isaac Newton "*Nature does nothing in vain when less will serve; for Nature is pleased with simplicity and affects not the pomp of superfluous causes.*" [Newton and Cajori, 2022] I will like to thank Dr Jason Lotay for his generosity with time and patience, Dr Derek Moulton for discussions regarding particular ideas, and Dr Marcos Alvarez for his brilliant physical insight.

References

- [Buchanan, 2023] Buchanan, M. (2023). On the wings of a butterfly. *Nature Physics*, 19(12):1750–1750.

- [Cahn and Hilliard, 1959] Cahn, J. and Hilliard, J. (1959). Free energy of a nonuniform system. *J. Chem. Phys.*
- [Dallmann and Herman, 1978] Dallmann, S. H. and Herman, W. S. (1978). Hormonal regulation of hemolymph lipid concentration in the monarch butterfly, *Danaus plexippus*. *General and comparative endocrinology*, 36(1):142–150.
- [Goriely, 2017] Goriely, A. (2017). *The mathematics and mechanics of biological growth*. Springer.
- [Huang et al., 2023] Huang, H., He, W., Zou, Y., and Fu, Q. (2023). Ustbutterfly: a servo-driven biomimetic robotic butterfly. *IEEE Transactions on Industrial Electronics*, 71(2):1758–1767.
- [Jones et al., 2023] Jones, A., Leary, M., Bateman, S., and Easton, M. (2023). Investigating mechanical properties of additively manufactured multimaterial gyroids: The effect of proportion, scale and shape. *Additive Manufacturing*, 76:103784.
- [Kaczmariski et al., 2024] Kaczmariski, B., Leanza, S., Zhao, R., Kuhl, E., Moulton, D. E., and Goriely, A. (2024). Minimal design of the elephant trunk as an active filament. *Physical Review Letters*, 132(24):248402.
- [Kammer, 1967] Kammer, A. E. (1967). Muscle activity during flight in some large lepidoptera. *Journal of Experimental Biology*, 47(2):277–295.
- [Newton and Cajori, 2022] Newton, I. and Cajori, F. (2022). *Sir Isaac Newton’s mathematical principles of natural philosophy and his system of the world*. Univ of California Press.
- [Pacard and Ritoré, 2003] Pacard, F. and Ritoré, M. (2003). From constant mean curvature hypersurfaces to the gradient theory of phase transitions. *Journal of Differential Geometry*, 64(3):359–423.
- [Pipkin, 1968] Pipkin, A. C. (1968). Integration of an equation in membrane theory. *Zeitschrift für angewandte Mathematik und Physik ZAMP*, 19(5):818–819.
- [Shames, 2017] Shames, I. H. (2017). *Energy and finite element methods in structural mechanics: SI units*. Routledge.
- [Srygley, 2007] Srygley, R. B. (2007). Evolution of the wave: aerodynamic and aposematic functions of butterfly wing motion. *Proceedings of the Royal Society B: Biological Sciences*, 274(1612):913–917.
- [Wilts et al., 2017] Wilts, B. D., Apeleo Zubiri, B., Klatt, M. A., Butz, B., Fischer, M. G., Kelly, S. T., Spiecker, E., Steiner, U., and Schröder-Turk, G. E. (2017). Butterfly gyroid nanostructures as a time-frozen glimpse of intracellular membrane development. *Science advances*, 3(4):e1603119.